Morgan Holland

Email: morgansholland72@gmail.com Cell Phone: (201) 602-2330 Linkedin: Morgansholland

Education

Rochester Institute of Technology
M.S. in Applied and Computational Mathematics, GPA: 3.27 August 2022 - May 2023
B.S. in Applied Mathematics and B.S. in Criminal Justice, GPA: 3.72 August 2018 - May 2023

Technical Skills

- **Programming Languages & Tools:** MATLAB, Python, SQL, JMP (SAS Software), Minitab, Excel
- **Data Analysis & Statistical Techniques:** Mathematical and Statistical Programming, Regression Analysis, Numerical Analysis, Machine Learning
- **Research & Modeling:** Mathematical Modeling, Data Analysis, Research

Professional Experience

Associate Data Analyst

GlobalLogic | Remote
July 2024 - Current

- Trained and tested Large Language Models (LLMs) for a Fortune 10 technology company, with a specialization in mathematics.
- Analyzed data to provide actionable insights and recommendations for optimizing model performance.
- Collaborated with cross-functional teams to implement improvements and achieve project milestones.

Statistician and Data Analyst - COVID-19 and Air Quality Research

Rochester Institute of Technology (RIT) | Rochester, NY
August 2022 – May 2023

- Analyzed COVID-19 intensities and case trends using JMP and Excel to assist in health protocol decisions.
- Utilized air quality sensor data to recommend optimal placement of air purifiers and ionizers.
- Partnered with RIT and Monroe County policymakers to provide data-driven insights.

Statistician and Data Analyst

Rochester Institute of Technology (RIT) | Rochester, NY
September 2021 – August 2022

- Conducted analysis on data from 170 criminal cases in Rochester using JMP, Minitab, and Excel.
- Developed statistical models to provide insights into crime trends.
- Performed kappa tests and regression analysis, presenting findings in comprehensive reports.

Academic Projects and Publications

Master’s Thesis: A Numerical Investigation into the Effect of the Immune System on Metastatic Breast Cancer

Rochester Institute of Technology (RIT) | January 2023 – May 2023

- Developed a coupled partial differential equations (PDE) model in MATLAB to analyze the immune system’s effect on cancer spread.
- Conducted simulations under various conditions to derive insights into cancer progression.

Publication: On The Power Series Solution to The Nonlinear Pendulum

Rochester Institute of Technology (RIT) | August 2020 – September 2022

- Applied asymptotic approximants to derive an exact infinite power series for a simple nonlinear pendulum.
- **Citation:** W. C. Reinberger, M. S. Holland, et al. https://doi.org/10.1093/qjmam/hbac013

Publication: Analytic Solution of the SEIR Epidemic Model via Asymptotic Approximant

Rochester Institute of Technology (RIT) | June 2020 – October 2020

- Utilized the SEIR model to predict COVID-19 infection spread on a global scale.
- Conducted analysis and identified key coefficients for equations.
- **Citation:** Steven J. Weinstein, Morgan S. Holland, et al. https://doi.org/10.1016/j.physd.2020.132633